

Roll No.

TCS-701

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

OBJECT ORIENTED ANALYSIS AND DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain various concepts used in object oriented paradigm with suitable diagram wherever required in detail. (CO1)
- (b) Draw a sequence diagram for withdrawing money from ATM. (CO3)
- (c) A retail system will interact with customers who place and track orders. In turn, the system will validate the user for ship orders and bill the customer. If all orders are not ready, part of the Order can be dispatched. Draw the Use case diagram and consider the following things : (CO3)

P. T. O.

- Identify the actors that interact with the element. Candidate actors include groups that require certain behavior to perform their tasks or that are needed directly or indirectly to perform the element's functions.
 - Organize actors by identifying general and more specialized roles.
 - For each actor, consider the primary ways in which that actor interacts with the element.
 - Consider also interactions that change the state of the element or its environment or that involve a response to some event.
 - Consider also the exceptional ways in which each actor interacts with the element.
 - Organize these behaviors as use cases, applying include and extend relationships to factor common behavior and distinguish exceptional behavior.
2. (a) Explain Object Oriented Software Development : A Use Case Derive approach in detail with suitable diagram. (CO2)
- (b) What is OOAD ? What are the benefits of OOAD over traditional analysis and design process of software development ? (CO2)
- (c) Draw a use case diagram for booking an online movie ticket booking system having the following functionalities : (CO3)
- A distinction is made between uses and registered users, only registered users can buy tickets. A user becomes a registered user by logging in. Unregistered users can register with a username that has not been taken by another user, an email address and a freely chosen password.

- Every user can search for movies, actors, directors and other persons and theatres.
 - For every movie and person there is a page showing their details. The movie details consists of a title, the year it was released in, a description, the genre of this movie and a photo. Also there are lists for the persons who were involved in this film, a soundtrack and some other information. The details to a person are restricted to its name, birthday, place of birth and a photo and the details of a theatre are composed of an address, show rooms and performances. Registered users can comment and rate movies, store their favourite movies and theatres in a personal list and buy tickets.
3. (a) Define an axiom. Mention the design axioms applied to object oriented design process with suitable example. (CO4)
- (b) What is the difference between activity diagram and sequence diagram ? Explain with example. (CO3)
- (c) A cellular network will interact with customers who place a call and receive a call. He/she can use scheduler to schedule the Phone Calls. There may be some exceptional behaviors of the users such as : a customer can also place conference call and can receive addition calls. Draw Use case diagram and consider following things : (CO4)
- Identify the actors that interact with the element.
 - For each actor, consider the primary ways in, which that actor interacts with the element.

(4)

- Consider interactions that change the state of the element or its environment or that involve a response to some event.
- Consider the exceptional ways in which each actor interacts with the element.
- Organize the behaviors as use cases, applying include and extend relationships to factor common behavior and distinguish exceptional behavior.

4. (a) Explain the Object Modeling Technique (OMT) presented by Jim Rumbaugh et al. (CO2)

(b) Model with a class diagram the following System : Vending Machine. A vending machine sells small, packaged, ready to eat items (chocolate bars, cookies, candies, etc). Each item has a price and a name. A customer can buy an item, using a smart card (issued by the vending machine company) to pay for it. No other payment forms (i.e. cash, credit card) are allowed. The smart card records on it the amount of money available. The functions supported by the system are :

- Sell an item (choose from a list of items, pay item, distribute item)
- Recharge the machine
- Set up the machine (define items sold and price of items)
- Monitor the machine (number of items sold, number of items sold per type, total revenue)
- Check the availability of an item

The system can be used by a customer, a maintenance employee (who check the items and recharges items in the machines), an administrator (who sets up the machine). However, We don't need to keep track of which user bought something or to keep track of the id or other details of his smart card. For example : We're not interested in saying that the customer Arushi bought 10 candies with the smart card # 1. We are interested in Vending machine sold 10 candies, each with price ₹ 1/.

(CO4)

- (c) What do you mean by a test plan and a test case ? Explain various types of testing strategies.

(CO5)

5. (a) List the methods of micro development process of Booch Methodology. Explain each method.

(CO2)

- (b) What are persistent and non-persistent objects ? Differentiate between them with the help of an example.

(CO4)

- (c) Explain LIML history in detail. What are the commercial applications of UML ?

(CO3)

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TCS-703

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

COMPUTER NETWORKS-II

Time : Three Hours

Maximum Marks : 100

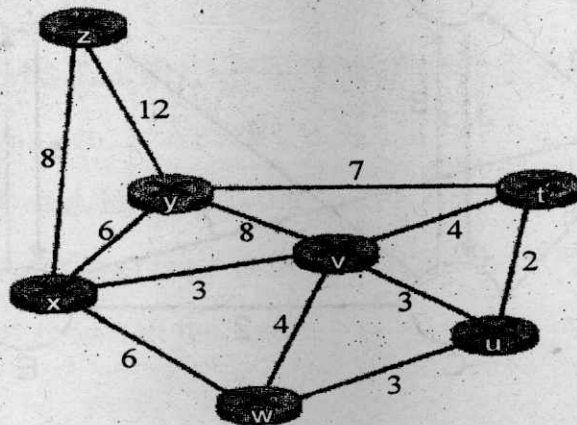
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

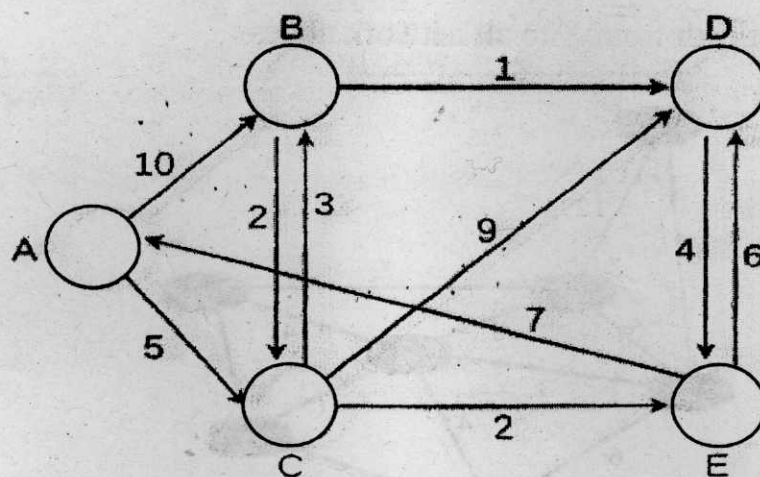
(iv) Each sub-question carries 10 marks.

1. (a) How does link state routing differ from distance vector routing ?
Consider the network given in Figure. With the indicated link costs, use Dijkstra's shortest-path algorithm used in link state routing to compute the shortest path from X to all network nodes : (CO1)

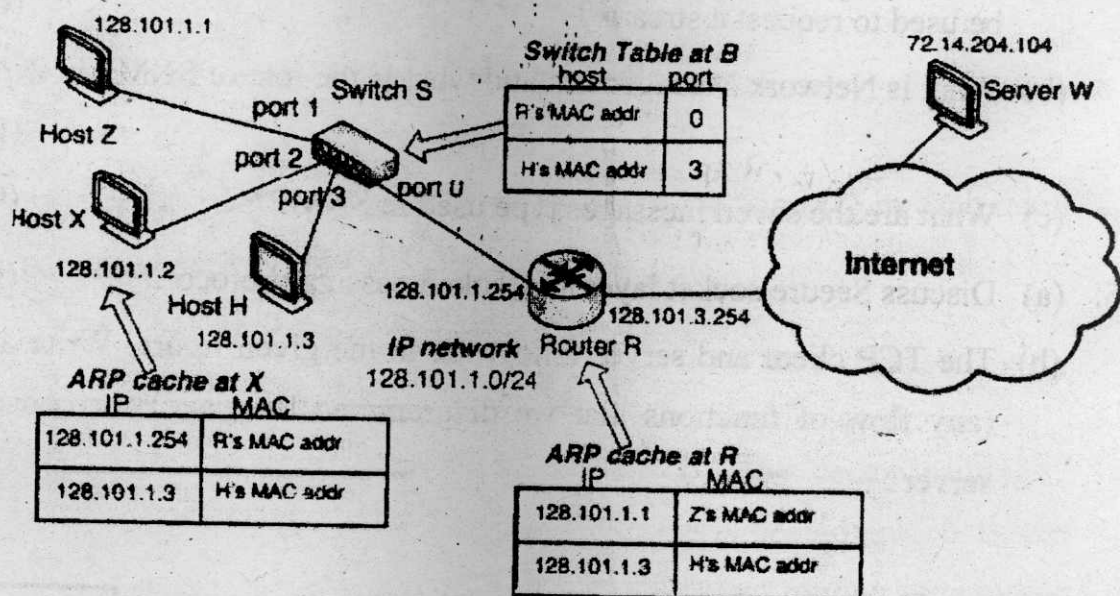


P. T. O.

- (b) (i) Comparison of LS and DV with robustness as the criteria.
 (ii) Discuss services offered by data link layer to network layer. (CO1)
- (c) Consider the three node topology, the link cost are $c(x, y) = 4$, $c(y, z) = 5$, $c(z, x) = 2$, compute the distance tables after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm. (CO1)
2. (a) Explain about Address resolution protocol Packet format : (CO2)
- (i) Encapsulation
 (ii) Proxy ARP
- (b) Explain how OSPF and RIP works as intra-domain routing protocols for autonomous systems. (CO2)
- (c) Write Distance Vector Algorithm. Consider the below network topology where circles represent routers and the routers are using link-state routing algorithm to determine the least cost routes to all other routers. Determine the routing table entries for routers B, C, D, and E after convergence. (CO2)



3. (a) What is the remainder obtained by dividing $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$ and explain correcting single bit error using single bit parity ? (CO3)
- (b) Consider the following switched network in the figure below, where we have one Ethernet switch S, connecting three hosts, host Z on port 1, host X on port 2 and host H on port 3, as well as an IP router R (default router for the host) on port 0. These hosts lie on an IP network with the



network prefix 128.101.1.0/24. The IP address for the interface of router R that is connected to switch S is 128.101.1.254. The IP addresses for hosts Z, X and H are shown in the figure. Furthermore, the current switch (forwarding) table at switch S and the ARP caches at host X and router R are shown in the figure.

P. T. O.

Suppose host X wants to send an IP datagram to host Z, and assume that host X knows the IP address of host Z (eg. Via DNS lookup).

Briefly explain how host X obtains the MAC address of host Z. Describe how switch S handles the ARP request and response messages, and builds its switch table. (CO3)

(c) Discuss RTP packet header fields. (CO3)

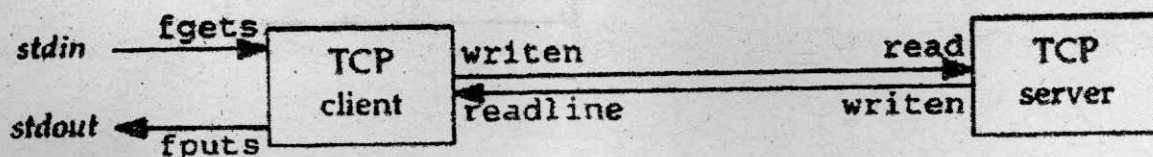
4. (a) How is RTSP similar to HTTP ? Does RTSP have methods ? Can HTTP be used to request a stream ? (CO4)

(b) What is Network Management and what is the role of SNMP in it ? (CO4)

(c) What are the seven messages type used in SNMP ? (CO4)

5. (a) Discuss Secure socket layer and Token passing protocol. (CO5)

(b) The TCP client and server follow the in the given figure. Write a code (any flow of functions that we diagrammed language) for concurrent server : (CO5)



(c) What is Socket and what does it consist of ? Explain raise condition.

(CO5)

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TCS-704

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the Myopic view of Computer Architecture. (CO1)
- (b) Discuss various Performance Metrics and its evaluation criteria in computer architecture. Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm². Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm². Find the yield and cost for both wafers. (CO3)
- (c) State any three of the following : (CO1)
 - (i) Iron Law of Performance
 - (ii) Amdahl's Law
 - (iii) Lhadmas Law
 - (iv) Moore's Law

P. T. O.

2. (a) Explain the various coherence and locality properties in detail.
(CO2)

(b) Formulate the set-associative cache memory mapping technique. A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB, compute the number of tag bits.
(CO3)

(c) Explain the concept of Virtual Memory. In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 4 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. What is the maximum number of bits that can be used for storing protection and other information in each page table entry?
(CO2, CO4)

3. (a) A four-stage pipeline has the stage delays as 150, 120, 160 and 140 ns respectively. Registers are used between the stages and have a delay of 5 ns each. Assuming constant clocking rate, what is the total time taken to process 1000 data items on the pipeline. Also explain in what circumstances, we can enhance the performance of the pipelining.

(CO5, CO6)

(b) Formulate the speedup, throughput, and efficiency of a pipelining system.
(CO2, CO3)

(3)

- (c) (i) List pipelining hazards with help of suitable example. (CO3)
- (ii) Consider the following reservation table for a pipeline having three stages S_1 , S_2 and S_3 . Compute the minimal average latency (MAL). (CO4)

Time →

	1	2	3	4	5
S_1	x				x
S_2		x		x	
S_3			x		

- 4 (a) Write short notes on the following : (CO4)
- (i) Branch Prediction
- (ii) Direction Predictor
- (iii) Hierarchical Predictors
- (b) Compare and contrast instruction level parallelism (ILP) and data level parallelism (DLP). (CO5)
- (c) Explain the duplicating register value policies in detail. (CO5)
5. (a) Classify the multi-processing computer architecture in context of parallel architectures. (CO5)
- (b) Differentiate the centralized and distributed shared memory in detail. (CO3, CO5)
- (c) List various issues and their solution in loosely coupled and tightly coupled systems. (CO5, CO6)

27.12.2022

Roll No.

TCS-706

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

ARTIFICIAL INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How to define a problem as state space search ? Discuss it with the help of an example. Give an explanation on the difference between strong AI and weak AI. (CO1)
- (b) Explain the Filmore's case grammar with help of sentence "Ram opened the door with the key" and "The door was opened by Ram with the key". (CO1)
- (c) Define Horn clause, goal clause, dual Horn clause, Horn formula, definite clause. (CO2)

P. T. O.

2. (a) Check the validity of the following implications : (CO3)

$P \rightarrow (Q \rightarrow R)$ equivalent to $(F \rightarrow Q) \rightarrow (P \rightarrow R)$

- (b) Parse the sentence "Kathy jumped the horse" using top-down and bottom-up parsing. (CO3)

- (c) Convert into first order predicate logic : (CO3)

(i) Marry loves everyone.

(ii) No one talks.

(iii) Everyone loves himself.

(iv) Everyone loves everyone.

(v) Everyone loves everyone except himself.

(vi) Every student smiles.

3. (a) What is Prolog in AI ? What does a production rule consist of ? (CO4)

- (b) Derive the parse tree for the following sentence by making use of an appropriate grammar. "Rama killed the demon". Productions rules are :

(CO5)

$S \rightarrow NP VP$

$NP \rightarrow N$

$NP \rightarrow DET N$

$VP \rightarrow V NP$

- (c) What is a pattern ? How is learning of pattern recognition through clustering done ? (CO6)

(3)

4. (a) What are the applications of Expert Systems in AI ? Explain DENDRAL, MYCIN. (CO5)
- (b) Prove the following : (CO3)
- (i) $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$
- (ii) $\neg(P \leftrightarrow Q) \leftrightarrow (P \leftrightarrow \neg Q)$
- (iii) $((P \rightarrow Q) \wedge \neg Q) \rightarrow (\neg P)$
- (c) What is correspondence between intelligent agents and human agent ? Explain with proper diagram. Difference between AI and Expert System. (CO3)
5. (a) What are the different steps to solve an AI problem ? Explain. What is a production system ? Explain with an example. (CO4)
- (b) Write a prolog program to implement power (X, N, V) where V equals X^n . (CO5)
- (c) What is expert system in AI ? What are the parts of an expert system and how do they work ? (CO5)

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TCS-710

**B. TECH. (CSE) SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

STORAGE NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Illustrate Information Life Cycle. What are the benefits of ILM ? (CO1)
(b) Explain the core elements of a hospital's data center. (CO1)
(c) Define server centric IT architecture and storage centric IT architecture with advantages and limitations. (CO1)
2. (a) Illustrate RAID 3. List out its benefits. Why is RAID 1 is not a substitute for a backup ? (CO2)
(b) What are different factors affecting disk drive performance ? (CO2)
(c) How can a block level virtualization implementation can be used as a data migration tool ? Explain how data migration will be accomplished and discuss the advantages of using this method for storage. Compare this method to traditional migration methods. (CO2)

P. T. O.

(2)

3. (a) List the factors contributed to the growth of digital data. (CO3)
(b) What is Striping ? Why is RAID 0 not an option for data protection and high availability ? (CO3)
(c) What is IP SAN ? Discuss the data storage applications that could benefit from an IPSAN solution. (CO3)
4. (a) Illustrate the elements, connectivity and management of Network Attached Storage (NAS) . (CO4)
(b) Why isn't Fibre Channel used often for the internal drives inside system cabinets ? List the *four* interconnects from slowest to fastest performance for storage networking applications. (CO4)
(c) Write a note on Hard Zoning, Soft Zoning and Hot spare. (CO4)
5. (a) What is DAS ? Give an example of when DAS is a good solution. (CO5)
(b) Illustrate Fibre Channel (FC) interconnectivity options that support FC architecture. Discuss the advantages of FC-SW over FC-AL. (CO5)
(c) A large company is considering a storage infrastructure—one that is scalable and provides high availability. More importantly, the company also needs performance for its mission-critical applications. Which storage topology would you recommend (SAN, NAS, IPSAN) and why ? (CO5)

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TCS-711

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

SPARK FUNDAMENTALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Can we deploy Spark application without Hadoop ? Justify if yes/no.

(CO2)

(b) Distinguish between the following :

(CO2)

(i) Spark Context and Spark Session

(ii) Transformations vs. Actions in RDD

(c) What are the benefits of using Dataframes in Spark ? Write in its core operations.

(CO2)

P. T. O.

2. (a) Write possible windows aggregation functions supported by Spark.
Explain each type of role in building a big data solution. (CO1)
- (b) It is said, 'StructField describes a single field in a StructType'. Justify this statement by taking some example. (CO1)
- (c) Why do we need literals when it is possible to add a new column easily in spark ? (CO1)
3. (a) How Apache Spark adapts Hive in real time ? (CO4)
- (b) Lets say we have a log file wherein we are required to work on set of operations like extraction, manipulation, substitution. Can Apache Spark be used for the above use case ? How ? (CO4)
- (c) (i) Write a Spark program to check a given list is a palindrome or not.
(ii) Write a Spark program to count the number of occurrences of each element in a csv. (CO4)
4. (a) Categorize different variety of data analysis supported by Apache Spark. (CO3)
- (b) Access Spark joins a distinguish distinct joins present in Spark SQL ? (CO3)
- (c) Measure the importance of Spark SQL in real world. (CO3)

(3)

5. (a) Data aggregation helps any organization to achieve specific business objectives. How Apache Spark supports data aggregation ? Write in about its capabilities. (CO5)
- (b) Write a Spark program and create two lists one containing all the odd numbers and other showing all the even numbers between 100 and 150. As a tip, there is no need to manually create a data structure to represent all the number literals, such as, 100, 101, 102, etc. Instead, use the handy Range type to help you create number literals between 100 and 150. (CO5)
- (c) Write a short note on Structured Collection in Spark. (CO5)

Roll No.

TCS-726

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
ARTIFICIAL INTELLIGENCE AND PATTERN
RECOGNITION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive a parse tree for the sentence "Bose loves the fish", where the following rules are used : (CO1)

$S \rightarrow NP VP$

$NP \rightarrow N$

$NP \rightarrow DET N$

$VP \rightarrow V NP$

$DET \rightarrow the$

$V \rightarrow loves$

$N \rightarrow Bose/fish$

P. T. O.

- (b) Define the following : (CO1)
agent, agent function, agent program, rationality and reflex agent.
- (c) Write a function `conv(C, F)` in prolog program to convert centigrade temperature to Fahrenheit. (CO1)
2. (a) Discuss the relative merits of depth-first and breadth-first searching methods. Suggest some application to which each is best suited. (CO2)
- (b) What are language models ? Assume that you want to provide a solution for language translation. Which natural language processing techniques you want to implement ? (CO2)
- (c) Write a prolog program that answer questions about family member and relationship. Include facts and rules which define sister, brother, father, mother, grandchild and grandfather. (CO2)
3. (a) Propose the architecture of neural network for classifying the samples into three classes named as A, B and C. Assume the dataset contains 50,000 images. How many samples do you want to use for training and validation ? Give reason for your answer. (CO3)
- (b) A technique for determining whether a computer could or could not demonstrate the artificial Intelligence, Presently, this technique is called Turing Test. Explain. (CO3)
- (c) Discuss differences between data mining and knowledge discovery in databases. (CO3)
4. (a) Define expert system. With the help of a block diagram, explain the operation of an expert system. (CO4)

(3)

- (b) What is a pattern, how learning of pattern recognition through clustering is done ? How does pattern recognition work ? What are patterns in terms of data ? (CO4)
- (c) Discuss nearest neighbour classification with a suitable example. (CO4)
- 5. (a) What is N-gram and POS (part of speech) ? If P is true and Q is false then P implies Q is and if P is false and Q is false then P biconditional Q is. Explain with help of truth table. (CO4 and CO5)
- (b) What is Hierarchical Clustering in Machine Learning ? (CO4 and CO5)
- (c) Write short notes on the following : (CO4 and CO5)
 - (i) DENDRAL
 - (ii) MYCIN.

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TCS-731

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

DIGITAL FORENSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Evaluate the significance of any *two* of the following and write short notes on the following : (CO3)

(i) Types of investigation software

(ii) Data hiding techniques

(iii) Anti-forensics

(b) Analyze the advantages and disadvantages of adopting digital forensics as a field of investigation. (CO1)

P. T. O.

- (c) Write short note on any *two* of the following : (CO5)
- (i) Multimedia forensics
 - (ii) Virtual Machine forensics
 - (iii) Database forensics
 - (iv) Network forensics
2. (a) What is the current state of IT laws in India and the World ? How can this landscape improve ? Give your recommendations. (CO1)
- (b) Discuss the evolution of forensic science as a field of study and evaluate the significance, role and key milestones in developing digital forensics as a field of study. Also present the key challenges faced this area in today's world. (CO2)
- (c) What are the challenges faced by a forensic investigator while acquiring evidence ? (CO4)
3. (a) Write a short note on any *two* of the following : (CO2)
- (i) Evidence validation
 - (ii) Cloud forensics
 - (iii) IoT forensics
 - (iv) Live forensics
- (b) What are the traditional problems associated with computer forensics ? Why is it important for computer security professionals to become familiar with digital evidence ? (CO3)
- (c) What are the different methods of password cracking used by forensic investigators ? Discuss the advantages and disadvantages of these methods. (CO5)

4. (a) Illustrate the digital forensic investigation model. Discuss each of the steps involved while investigating a digital forensic case. (CO2)
- (b) What are the various tools used in a forensic investigation? How would you investigate an online financial fraud case? (CO3)
- (c) Articulate how cyber terrorism and information warfare are affecting the state of ICT. (CO5)
5. (a) If you are positioned as a cyber attacker, what techniques/tools/procedures will you use to compromise a resource? (Feel free to make assumptions and present a case study on the same). (CO1)
- (b) A disgruntled user was caught copying some files from hard disk to pen drive. When someone saw him, he quickly forcefully deleted data from pen drive and hard disk. Computer is having Windows 10 installed in it. This has been reported to incident handling team, first responder came there to collect evidence from that running computer. The system might contain something crucial as an evidence in system like open files or some hidden files or some files on RAM. Write all tools and steps you will take to collect digital evidence from hard disk and pen drive. Please note that integrity of the data must be maintain. Log all hashes from start till end. (CO4)
- (c) Evaluate the significance and write a short note on IT laws in India including the Digital Personal Data Protection Bill, 2022. (CO5)

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TCS-733

**B. TECH. (CS) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DATA SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the specific skills required for Data Scientist ? (CO1, CO2)
(b) Define any *four* out of six data analysis phases. (CO1, CO2)
(c) Explain complex data type in R, using suitable example. (CO1, CO2)
2. (a) Demonstrate creation and calling a function in R, using a suitable example. (CO1, CO2, CO3)
(b) Differentiate between discrete and continuous variable. (CO1, CO2, CO3)
(c) Explain the different types of missing values. (CO1, CO2, CO3)
3. (a) Explain the four "V" for Big Data. (CO1, CO3)

P. T. O.

(2)

- (b) How many phases are there in Data Life Cycle ? Explain each of them in brief. (CO1, CO3)
- (c) Differentiate between primary data and secondary data with suitable example. (CO1, CO3)
- 4. (a) What is data anonymization ? What types of data should be anonymized ? (CO1, CO4, CO6)
- (b) Write and explain the code for generating "bar graph" using ggplot2 library. (CO1, CO4, CO6)
- (c) Differentiate between classification and regression techniques. (CO1, CO4, CO6)
- 5. (a) Explain type1 and type2 error with suitable example. (CO3, CO5)
- (b) Write down the importance of outliers in statistical analysis. (CO3, CO5)
- (c) Explain standard deviation and interquartile range and write R code to compute standard deviation and interquartile range. (CO3, CO5)

Roll No.

TCS-750

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

CLOUD ORCHESTRATION AND LOAD BALANCING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Midtrans and AWS collaborate to develop Pojok Usaha—a centralized online portal featuring over 30 products and services—to drive digital transformation for Indonesian SMBs and simplify application and procurement of cloud services. Midtrans provides an e-payment gateway platform used by more than 500,000 businesses. Evaluate cloud automation with suitable case study of above mentioned organization. (CO1)

P. T. O.

(b) Content analytics provider smartocto accelerated its time to market and deployed a predictive editorial analytics solution. Smartify in less than 3 months by using Amazon SageMaker. Evaluate Grid Computing with suitable case study of above mentioned organization. (CO1)

(c) BNS Group reduced infrastructure costs by 50 percent and gained access to the AWS ISV sales and client network by migrating to AWS. BNS Group is an Australian ISV offering enterprise SMS and fax server solutions to public and private sector clients. Evaluate key scheduling technique with suitable case study of above mentioned organization.

(CO1)

2. (a) Lyell Immunopharma is automating GxP compliance for its environments and applications with Rapid Q, an automated validation solution built on AWS. Evaluate Amazon Web Compute Services with suitable challenges of above mentioned organization. (CO2)

(b) Sterling Auxiliaries is an international manufacturer of surfactants and industrial chemicals based in India. To meet its go-live timeline and improve system performance, the company upgraded its on-premises SAP R/3 system to SAP S/4HANA on AWS. Evaluate Google App Engine Cost Model deploy possibility with suitable case study of above mentioned organization. (CO2)

(c) India-based ALTBalaji is a subscription-based video on demand (SVOD) platform that produces original over-the-top (OTT) media content. To broadcast live streams of its Indian reality show Lock Upp, the company chose to build its live streaming infrastructure on AWS. Evaluate cloud metering technique with suitable case study of above mentioned organization. (CO2)

3. (a) Space technology company Axelspace built an event-based, custom data pipeline on AWS, helping it optimize costs, free its development teams from infrastructure maintenance, and deliver data faster. Evaluate Ingram Micro Cloud Orchestrator with suitable case study of above mentioned organization. (CO3)
- (b) Veolia Australia and New Zealand migrated 34 applications to AWS to improve agility in development and project delivery, while integrating data for higher visibility across the business. Evaluate Microsoft Cycle Computing services compatibility with suitable case study of above mentioned organization. (CO3)
- (c) Technology company Bazaarvoice reduced machine learning inference costs by 82 percent and accelerated content delivery by using the ML capabilities of Amazon SageMaker Serverless Inference. Evaluate IBM Cloud Orchestrator Services with suitable case study of above mentioned organization. (CO3)
4. (a) Tokenize Xchange (Tokenize) is a Singapore-based digital exchange platform that facilitates the trading of over 100 crypto currencies. Tokenize is fully built and deployed on Amazon Web Services (AWS) and automatically scales storage and compute capacities based on transaction volumes and active concurrent users. Evaluate MapReduce deployment challenges of above mentioned organization. (CO4)
- (b) Discover how the US Army TATRC worked with Deloitte to create a cloud-based telehealth application that connects patients and clinicians with a push of a button. Evaluate spark deployment challenges of above mentioned organization. (CO4)

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- (c) HBO Max, Warner Media's direct-to-consumer platform offering best-in-class quality entertainment, uses Amazon Web Services (AWS) to scale. Evaluate GraphLab deployment challenges of above mentioned organization. (CO4)

5. (a) Baker Hughes migrated its computational fluid dynamics applications to AWS, cutting gas turbine design cycle time, saving 40 percent on HPC costs, and reducing its carbon footprint by 99 percent. Evaluate REST interface deployment in above mentioned organization. (CO5/CO6)

- (b) Learn from travel and hospitality companies including Priceline, Truventia, and Best Western that are using modern contact center solutions built on Amazon Web Services (AWS). Evaluate AWS load generator implementation process with suitable case study of above mentioned organization. (CO5/CO6)

- (c) Credit safe migrated on-premises systems to AWS- to improve scalability, reliability, and availability. This has eliminated years of technical debt and improved ways of working while eliminating risk. Evaluate MapReduce deployment programming model challenges with suitable case study in above mentioned organization. (CO5/CO6)

Roll No.

TCS-751

**B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

CLOUD SECURITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the motivation factors for cloud security ? Mention each of them in detail. (CO1)
- (b) Analyze the security management with all the factors in Cloud Security. (CO1)
- (c) Discuss the possible level of threats in infrastructure security at the network level, host level, and application level. (CO2)
2. (a) Discuss the different security measures required for deploying applications at IAAS, PAAS, and SAAS. (CO2)

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(2)

- (b) Discuss the security aspects of the different layers of cloud computing in detail. (CO3)
- (c) Analyze the security issues at the virtualization level in the cloud and detail each of them. (CO3)
- 3. (a) Analyze the threat modeling approaches for securing cloud computing. (CO4)
- (b) What do you mean by VM escape attack ? Explain it with its type and diagram. (CO4)
- (c) Analyze the cloud malware injection attack. Explain each of their detail. (CO4)
- 4. (a) Explain IAM challenges in detail with all its category. (CO5)
- (b) Describe the IAM functional architecture with its diagram. (CO5)
- (c) Describe all the protocols used in identity and access management. (CO5)
- 5. (a) Analyze the different types of IDS suitable for cloud computing security and explain each of them in detail. (CO6)
- (b) Analyze the different types of intrusion detection systems suitable for the cloud and mention each of them. (CO6)
- (c) Draw the block diagram of the conceptual working of VMI approaches and explain its functioning if possible. (CO6)

Roll No.

TCS-756

B. TECH. (CSE/CC+IS/CE/CST/SE/IT)

(SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022 .

HUMAN-COMPUTER INTERACTION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) For programming question provide typical output.

1. (a) Define HCI. Discuss the importance and advances in audio-visual Interface as a tool for improved productivity of Human User. (CO1)

(b) Explain the role of Engineering, Computer Science, Psychology, and design in the development and advances of HCI. (CO2)

(c) What can a system designer do to minimize the memory load of the user? (CO1)

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2. (a) What are the steps of UI Design and Development Process ? Explain with useful figure. (CO1)
- (b) Discuss the typical-computer interaction (HCI) models, styles, and various historic HCI paradigms. (CO6)
- (c) Find out the natural language interfaces. Are there any successful systems ? For what applications are these most appropriate ? (CO3)
3. (a) Analyze and identify user models, user support, socio-organizational issues, and stakeholder requirements of HCI systems. (CO4)
- (b) Explain the user interface for the Computing Systems ? Explain how these interfaces have changed in the past. (CO2)
- (c) Design (in words as well as graphically) the interaction framework introduced in Human-Computer Interaction. Show how it can be used to explain problems in the dialog between a user and a computer. (CO6)
4. (a) What are VR and AR ? Discuss the role VR and AR in the development and advances of HCI. (CO1)
- (b) Apply an interactive design process and universal design principles to designing HCI systems. (CO3)
- (c) How can design rationale benefit interface design and why might it be rejected by design teams ? (CO6)

(3)

5. (a) Conclude the name is given to a hand-held pointing device that controls the position of the pointer on the screen with suitable example. (CO5)
- (b) Determine the challenges in speech recognition techniques in artificial intelligence with case study. (CO5)
- (c) Can you think of any instances in which the 'noun-verb' guideline for operations, as suggested in the Apple human interface guidelines for the Desktop Interface, would be violated ? Suggest other abstract guidelines or principles besides consistency which support your example. (CO6)

(Hint : Think about moving files around on the Desktop.)

Roll No.

TCS-761

B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
CLOUD INFRASTRUCTURE SERVICES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate between public cloud and private cloud (any *five* points).

(CO1)

(b) State the features of Cloud Service Models (any *five* points). (CO1)

(c) What fundamental advantages does cloud computing technology bring to scientific applications ? (CO1)

2. (a) (i) Elaborate the term "Software as a Service" related to cloud computing. 6 (CO2)

(ii) Give the drawbacks of Cloud Computing Paradigm. 4 (CO2)

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(2)

- (b) What are the suitable conditions of PaaS ? Discuss briefly. (CO2)
- (c) (i) Explain how to attain QoS by managing cloud ? (CO2)
- (ii) Explain Cloud Service Life Cycle in detail.
- 3. (a) What do you understand by Service Oriented Architecture (SOA) ? Explain how it supports cloud computing. (CO3)
- (b) Compare and contrast distributed internet architecture vs. SOA with respect to application processing and application logic. (CO3)
- (c) Sketch down the elements and relationship between service requester and service provider. (CO3)
- 4. (a). Cloud has three stages : Software as a Service (SaaS), Infrastructure as Service (IaaS), Platform as a Service (PaaS). Draw a pictorial diagram to explain these offered through the cloud environments and explain them with the help of suitable examples and name the popular companies who are offering these services. (CO4)
- (b) Write a brief note on Google's Cloud Storage. (CO4)
- (c) Discuss the Google megastore model with an example. (CO4)
- 5. (a) Briefly explains the architecture of IBM Smart Cloud with a neat architectural diagram. (CO5)
- (b) List out and brief the features of Amazon Simple Queue Service. (CO5)
- (c) Describe the working of Microsoft Azure with diagram. (CO5)

Roll No.

TCS-771

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

NATURAL LANGUAGE PROCESSING

WITH BIG DATA

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are Morphemes in Linguistics ? Give *two* examples that differentiate morphemes from words. (CO2)
- (b) Briefly describe any *two* modern techniques that are used as alternatives to bag of words (BoW) ? Mention any *two* advantages of the modern techniques over the BoW model. (CO1)

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- (c) Where is part of speech tagging useful in the area of natural language processing ? Explain using *two* suitable examples. (CO3)
2. (a) Using Python, write suitable code to create IF and TF-IDF weights of four sentences stored in string variables s1, s2, s3, s4. (CO1)
- (b) Briefly analyze any two linear classifiers used in NLP. How is a classifier's fit() method different from transform() ? (CO5)
- (c) (i) What do you understand by pretrained language models in deep learning ? Give examples of any *two* pretrained language models.
- (ii) In the context of deep learning techniques, explain model fine tuning with the help of a suitable example. (CO5)
3. (a) Differentiate between word2vec embedding and contextual embeddings. How do contextual embeddings improve NLP tasks ? Explain briefly. (CO5)
- (b) (i) What is BERT language model ?
- (ii) Explain language modelling and next sentence prediction tasks in BERT pretraining. (CO5)
- (c) (i) In terms of feature extraction, how do deep learning techniques differ from traditional techniques of natural language processing ?
- (ii) Briefly describe any two recent deep learning techniques (not from this paper) for natural language processing. (CO4)

(3)

4. (a) Using Naïve Bayes classifier, explain how you would classify Spam vs. Ham messages. Why is Laplace smoothing used in NB classifier ?

(CO5)

- (b) Out of 1000 samples, 600 are positive and 400 are negative. A classifier is used and it classifies 500 correct positive samples and 350 correct negative samples. Create the confusion matrix and calculate precision, recall and F1-score.

(CO5)

- (c) Explain the following problems in NLP :

(CO4)

(i) Curse of dimensionality

(ii) Data sparsity

5. (a) Explain the various steps involved in machine translation of text from one language to another using traditional techniques. Mention any *two* challenges involved in traditional machine translation.

(CO4)

- (b) Briefly describe two traditional (apart from Naïve Bayes) and two deep learning techniques for text classification.

(CO4)

- (c) In a chatbot system what do the terms Intents and Entities mean ? Mention any *four* challenges in creating a chatbot system.

(CO6)

Roll No.

TCS-722

**B. TECH. (IT) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

DATA WAREHOUSING AND DATA MINING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the various data reduction techniques used in data mining.

(CO2)

(b) Describe the architecture of data mining. What are the various classifications of data mining ?

(CO2)

(c) Explain the concept of Concept hierarchy generation with a suitable diagram.

(CO2)

2. (a) Explain the data cube aggregation approach for data mining along with a suitable example.

(CO2)

(b) Explain Apriori algorithm with a suitable real life example.

(CO2)

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(c) How do you mine multilevel association rules from transaction databases ? (CO2)

3. (a) Construct the decision tree using ID3 algorithm : (CO3)

Outlook	Temperature	Humidity	Windy	Golf ?
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

(b) Illustrate the necessary steps of Back propagation algorithm. (CO3)

(c) Explain the approach of density based clustering methods. Compare and contrast DBSCAN and OPTICS. (CO3)

(3)

4. (a) Define Data Warehouse. Illustrate the differences between Database System and Data Warehouse. (CO2)
- (b) Elaborate 3 tier architecture of a data warehouse with a suitable diagram. (CO2)
- (c) Explain the Data Cubes, Stars, Snow Flakes, Fact Constellations used in multidimensional data models. (CO2)
5. (a) In data warehouse technology, a multiple. dimensional view can be implemented by a relational database technique (ROLAP) by a multidimensional database technique (MOLAP), or by a hybrid database technique (HOLAP) . For each technique explain how each of the following functions may be implemented : (CO4)
 - (i) Generation of a data warehouse (including aggregation)
 - (ii) Roll-up
 - (iii) Drill-down
 - (iv) Incremental updating
- (b) How tuning is performed in Data Warehouse ? Why is it required ? Is there any implications in tuning process ? If yes, specify. (CO4)
- (c) Examine the impact of Security on Design. How is security incorporated in the data warehouse ? (CO4)

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TCS-781

B. TECH. (CSE-SE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

SOFTWARE QUALITY ASSURANCE AND RELIABILITY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the six broad, independent categories of quality characteristics ? (CO1, CO2, CO3)
- (b) Decompose the quality characteristics of Functionality, Reliability, and Usability. (CO1, CO2, CO3)
- (c) Point out the objectives of SQA activities. (CO1, CO2, CO3)
2. (a) Assume a 1000 function point IT project such as an accounting package and 53,300 Java statements for all cases. Assume 100 users of the software after deployment. Assume average complexity (problem, code,

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and data) in all four cases. The assumed test stages in the article include (1) unit test, (2) function test, (3) regression test, (4) component test, (5) usability test, (6) performance test, (7) system test, and (8) acceptance test.

If you don't use CMMI but have a good team and use Agile, Java, 10% reuse, and static analysis but no inspections and no certified test personnel.

- (i) Will the software delivered have high severity bugs ?
 - (ii) Will the software delivered have security flaws ?
 - (iii) Will the software delivered have error-prone modules ?
 - (iv) Is there a probability of "bad fixes" ?
 - (v) What can be the average customer satisfaction ? (CO1, CO2, CO3)
 - (b) Associate the costs involved with software maintenance quality activities. (CO1, CO2, CO3)
 - (c) Explain the formal design review process with the help of process diagram. (CO1, CO2, CO3)
3. (a) Write a note on the classification of the causes of software errors. (CO1, CO2, CO3)
- (b) With a diagram, interpret SQA architecture. (CO1, CO2, CO3)
 - (c) Point out the benefits and risks of involving external participants in the software maintenance process. (CO1, CO2, CO3)

4. (a) Reframe and depict the traditional development life cycle to CASE tool supported development life cycle. (CO4, CO5)
- (b) This product (herein referred to as Product A) is available on Web and iOS applications. Our team integrated into the development process, attending sprint planning meetings, daily standups and weekly scrums. Our team developed automated test suites using our in-house Selenium wrapper for both Web and iOS applications. Test suites cover all essential regression features as well as other automatable feature test. Product A iOS Mobile Acceptance Test Suite is configured to test against real devices or iOS simulators, however, due to limitations of the application's requirements, key functionality is not possible on simulators. The test suite thereby will trigger test runs on physical devices located on campus. The reports themselves do not provide a good high-level view of the user stories for which they are tested. The test suite configuration is also not clear, in addition to the standard automation of the product, should a detailed cost analysis and recommendation be performed and given with regards to the Google App Engine configuration of the product.
- (i) What are the actions the team should take regarding the simulator ?
- (ii) Should the team focus on generating a detailed report and documentation ? (CO4, CO5)
- (c) Write a note on the training and certification process that is followed to ensure quality. (CO4, CO5)

5. (a) Write a note on Maintenance Quality Metrics. (CO4, CO5)

(b) What are the steps involved in Software Process Assessment Cycle ?

(CO4, CO5)

(c) Assume that a software system is undergoing system-level tests and the initial failure intensity is 15 failures per CPU hour. The failure intensity decay parameter has been found to be 0.025 per failure. So far, the test engineers have observed 60 failures. What is the current failure intensity ? (CO4, CO5)

Roll No.

TCS-791

B. TECH. (CSE-IS/DS/ML/SE)

(SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

INTERNET/NETWORK SECURITY

Time : Three Hours

Maximum Marks : 100

Note :- (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Investigate the working mechanism of ransomware attacks. What are the different solutions available for ransomware attacks ? (CO1)
- (b) What is the use of classifying the IP addresses ? Explain the following types of IP addresses : (CO1)
unicast IP address, multicast IP address and broadcast IP address.
- (c) Describe following security requirements : (CO1)
Integrity, authentication and access control.

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2. (a) Demonstrate the working mechanism of SSL handshake protocol. Why SSL alert protocol is required ? (CO3)
- (b) How would you compare the encryption procedure of data encryption standard (DES) algorithm and 2-3 DES algorithm ? (CO3)
- (c) What are the uses of the Advanced Encryption Standard (AES) algorithm ? Investigate the working mechanism of the AES encryption process. (CO3)
3. (a) Demonstrate the mechanisms of digital signature generation and verification of Digital Signature Standard (DSS) ? (CO4)
- (b) Design a secure communication system in which you have to achieve different security properties, like, message authentication, data integrity checking and secrecy of the exchanged messages. Use hash, public key cryptography and symmetric key cryptographic-based operations in your answer. (CO4)
- (c) How would you compare the following security mechanisms, "authentication tied to plaintext" and "authentication tied to ciphertext" ? (CO4)
4. (a) Demonstrate the mechanisms of digital signature generation and verification of RSA digital signature algorithm. (CO5)
- (b) What are the uses of Virtual Private Network (VPN) ? Explain the working mechanism of the VPN using some figures. (CO5)
- (c) Why do we need a Virtual Private Network (VPN) tunnel ? How would you compare the following tunneling methods, "split tunneling" and "GRE tunneling (IP tunneling)" ? (CO5)

5. (a) Demonstrate the working mechanisms of following types of firewalls :

(CO6)

packet-filtering firewall, proxy firewall and next-generation firewall.

- (b) Investigate the working mechanisms of following malware programs :

Rootkit, trojan horse and worm.

(CO6)

- (c) Write down the differences between aggressive scan and decoy scan of NMAP tool. Also write down the syntax of aggressive scan and decoy scan.

(CO6)

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TIT-701

**B. TECH. (IT) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

MOBILE COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Describe WLAN-based architecture. How does a WLAN function ?
(CO1)
(b) Discuss about the purpose of GPRS. Discuss its benefits and goals.
(CO1)
(c) How do you compare hybrid with fixed channel allocation ? Which one would you prefer and why ?
(CO1)
2. (a) What is a cellular Network ? In context to cellular network, discuss the following :
(CO1, CO2)
Cluster, Re-use distance, cell splitting and sectorisation.

P. T. O.

(2)

- (b) What is meant by handoff interval and handoff region ? Explain their usefulness with appropriate diagrams. (CO1, CO2)
- (c) Why do we need WAR 2.0 ? On which Networking Model is WAP based upon ? (CO1, CO2)
- 3. (a) Discuss the issues which arises in data management of the mobile databases. (CO3)
- (b) Explain working principles of reconciliation and rumor replicated file system. (CO3)
- (c) What is Coda ? Why is coda promising and potentially very important ? (CO3)
- 4. (a) What are the requirements of mobile agents ? Discuss life cycle of mobile agent-based computing with suitable diagram. (CO4)
- (b) A mobile agent experiences the various events in its life cycle. Discuss in details. How the mobile agents are helpful in the reduction of the network load ? (CO4)
- (c) What are advantages and disadvantages of Mobile Agent ? Discuss some suitable applications of mobile agent briefly. (CO4)
- 5. (a) Discuss characteristics of Mobile Adhoc Networks. Why is it not possible to use circuit switching in Adhoc networks ? (CO5)
- (b) What are advantages and disadvantage of reactive protocols ? (CO5)
- (c) Explain about global state routing protocol with suitable used cases. (CO5)

Roll No.

TIT-704

**B. TECH. (CSE/IT) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Explain security architecture. Justify its relevance and significance in accomplishing authentication, authorization, confidentiality and integrity. (CO1)
(b) Which security services are guaranteed when using each of the following methods to send mail at post office ? (CO1)
(i) Regular mail
(ii) Regular mail with delivery confirmation
(iii) Regular mail with delivery and recipient signature

P. T. O.

- (c) Which technique, cryptography or steganography is used in the following cases for confidentiality? (CO1)
- (i) A student writes the answer to a test on a small piece of paper, rolls up the paper and inserts it in a ball point pen and passes the pen to another student.
 - (ii) To send a message a spy replaces each character in the message with a symbol that was agreed upon in advance as character's replacement.
 - (iii) A company uses special ink on its check to prevent forgery.
2. (a) Suppose that you have a block cipher where $n = 64$. If there are 10 1's in the cipher text, how many trial and error tests does Eva need to do to recover the plaintext from intercepted cipher text in each of the following cases? (CO2)
- (i) The cipher is designed as substitution cipher.
 - (ii) The cipher is designed as transposition cipher.
- (b) Illustrate the four transformations defined for AES. Also identify among these which one does not change the contents of the bytes? (CO2)
- (c) Compare the OFB and CTR mode of operation. (CO2)
3. (a) Users A and B use the Diffie-Hellman key exchange technique with a common prime $q = 7$ and a primitive root $\alpha = 5$. (CO3)
- (i) If user A has private key $X_A = 3$, what is A's public key Y_A ?
 - (ii) If user B has private key $X_B = 4$, what is B's public key Y_B ?

(3)

- (b) In an RSA cryptosystem, a particular user A uses two prime numbers $p = 3$ and $q = 11$ to generate her public and private keys. If the public key of A is $(7, n)$ and plaintext M is 35. Then find the private key of A and cipher text. (CO4)
- (c) Mention all the characteristics of a Hash function. Also discuss the processing of SHA-512. (CO4)
4. (a) Discuss the working of Kerberos in detail along with its limitations. (CO5)
- (b) Describe IPSEC's architecture and the services it offers in detail. (CO5)
- (c) What is the purpose of X.509 directory authentication services ? Also explain the general format of X.509 certificate. (CO5)
5. (a) Analyze the authentication schemes of IEEE 802.11. (CO6)
- (b) Categorize the cloud security attacks along with their countermeasures. (CO6)
- (c) List and briefly define the parameters that define an SSL-session state and an SSL session connection. (CO6)

Roll No.

TIT-721

B. TECH. (CSE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

BUSINESS INTELLIGENCE**Time : Three Hours****Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is your understanding of Business Analytics ? List few common core business functions (CO1, CO2)

(b) Discuss about leadership, strategic planning and customer focus process of Malcolm Baldrige Criteria for performance excellence framework.

(CO1, CO2)

(c) How can BI be used to enhance customer experiences ? Explain with an example. (CO1, CO2)

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2. (a) Can XML data be converted into a structured format ? Explain with an example. (CO1, CO2)
- (b) Can you think of an instance where you came across data that was stored or presented to you in an unstructured, semi-structured, and structured data format ? (CO1, CO2)
- (c) Web pages are said to be unstructured data even though they are defined by the HTML markup language which has a rich structure. Why ? (CO1, CO2)
3. (a) How is OLTP different from ERP systems and how is ERP different from BI ? (CO3, CO6)
- (b) Compare data warehouse and OLTP database. Is it important to have your data warehouse on a different system than your OLTP system ? Explain your answer. (CO3, CO6)
- (c) Which System (OLTP/OLAP) has support for concurrency control and recovery and why ? (CO3, CO6)
4. (a) How is the Ralph Kimball's approach to building a data warehouse different from the William H. Inman's approach to building a data warehouse ? Explain. (CO4)
- (b) What do you mean by Cross-Cell ? Explain with an example. (CO4)
- (c) How does maintain the data in a multidimensional format help with data analysis ? Comment. (CO4)

(3)

5. (a) Define KPISs to measure the effectiveness of a learning program.

(CO1, CO5, CO6)

(b) How does "measure" help with good decision making ? Explain.

(CO1, CO5, CO6)

(c) Why is it important for enterprises to go for "enterprise dashboard" ? Explain.

(CO1, CO5, CO6)